

Geophysical Research Letters®



RESEARCH LETTER

10.1029/2024GL108210

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Key Points:

- Rainfall intensity estimated from sound measurements in the Amazon rainforest, tipping bucket rain gauge, and machine learning models
- The best model successfully detects rainfall in 88% of the cases, with $R^2 > 0.87$ for hourly rainfall rates on the training site
- Model validated over two sites in the Amazon, 97% accuracy identifying rainfall events, R^2 of 0.69 and 0.93 for hourly rainfall rate

Supporting Information:

Supporting Information may be found in the online version of this article.

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Citation:

Xavier, R. S., Gosset, M., Maciel, T. F., Bicudo, T., do Nascimento, L. A., Ramalho, E., & Fleischmann, A. (2024). Measuring Amazon rainfall intensity with sound recorders. *Geophysical Research Letters*, 51, e2024GL108210. <https://doi.org/10.1029/2024GL108210>

Received 21 MAY 2024

Accepted 29 AUG 2024

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Measuring Amazon Rainfall Intensity With Sound Recorders

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Abstract Ground weather observations are scarce in many parts of the globe, hampering effective climate monitoring and disaster management. In the Amazon basin, this occurs due to its remoteness and the challenging measurement of rainfall within the forest. Innovative rainfall estimation methods are thus requested to fill this gap. Here we present an approach to estimate rainfall based on sound measurements. We identified the best frequency range to estimate rainfall occurrence and intensity, trained classification and regression models with sound and rain gauge data collected in the Central Amazon during 9 months. By training a random forest classifier/regression model based on power spectrum values it was possible to identify and satisfactorily estimate hourly rainfall rates in two vegetation environments distinct from the training site, located 30 km from it. The proposed method is a promising approach for future weather monitoring in remote tropical areas.

Plain Language Summary Understanding and predicting rainfall is a complex task, especially in areas where the availability of data from surface stations is limited, a common feature in many developing regions with insufficient rain gauge coverage. Recently, new opportunistic methods of rainfall measurement have emerged. Among them, is the use of the relationship between rainfall intensity and the sound produced by droplets hitting a surface. Sound recorders offer a low-cost solution and could provide an interesting means to increase spatial coverage of rainfall measurements, but also to fill information gaps under dense forests where conventional devices do not work. Our study developed a new technique and applied it to the Central Amazon region, by training a supervised machine learning model applied to sound recordings obtained in a tropical rainforest. To our knowledge, for the first time, such techniques are validated in locations far from the calibration site. We showed that reasonable results can be obtained for sites with distinct vegetation types and up to 30 km of distance from where the training data was acquired. Our findings demonstrate a strong capability for estimating hourly rainfall rates.

1. Introduction

Rainfall dynamics sustains ecosystems and supports human livelihoods around the world, shaping agricultural activities and freshwater availability. A dense surface network of rainfall ground measurements with long and continuous time series can provide essential information to our understanding regarding future global climate and its impacts on ecosystems and human society. However, the current world land climate monitoring infrastructure is highly concentrated in the Northern Hemisphere (WMO, 2023), where developed nations can afford to acquire and maintain a well-distributed network with high spatial resolution. Remote locations with lower economic resources have little to no observational data available. Therefore, it is necessary to develop solutions that enable the collection and dissemination of reliable climate data in such underrepresented regions. This is particularly relevant in the tropics, where rainfall is the most important meteorological variable due to its high spatial/temporal variability and connections with local atmospheric circulation (Figueroa & Nobre, 1990), agriculture, and hydrological cycle (Fassoni-Andrade et al., 2021). Recent increases in extreme events in the Amazon River basin also demand monitoring of rainfall rates at higher resolution than typically available to support the authorities in assisting local communities.

The Amazon biome is one of such remote and underrepresented regions, in terms of climate monitoring infrastructure. As the world's largest rainforest, it extends over nine countries and hosts approximately one-third of Earth's known species (Heckenberger et al., 2007). It plays a crucial role in regulating the rainfall cycle over the South American continent by transporting warm air and humidity through the low-level jet system along the Andes Mountain chain (Vera et al., 2006). In particular, most of the Brazilian northern region is located within the

Amazon biome, and although it represents 45% of the country's territory, it holds only 15% of the total number of automatic surface weather stations operated by the Brazilian National Institute of Meteorology (INMET, 2024). Notably, 37% of them were not operating by the time this manuscript was written. Most of these stations are positioned along rivers, and there exists a significant contrast in rainfall intensity near rivers compared to locations distant from water bodies, leading to observation biases regarding basin-scale total rainfall (Molion & Dallarosa, 1990; Paiva et al., 2011). New data are thus needed to compensate for these gaps in the current observational system.

Recently, innovative ideas to measure rainfall with non-conventional methods have emerged, including the estimation of meteorological variables based on telecommunication networks and the Internet of Things (Uijlenhoet et al., 2018). Such opportunistic methods provide perhaps the best way to solve the current lack of in situ observations in remote areas of the Amazon. An alternative and innovative technique for estimating rainfall that has been recently developed makes use of the soundscape of a given location, that is, its acoustic characteristics (Avanzato & Beritelli, 2020; Bedoya et al., 2017; Çoban et al., 2023; Döner et al., 2018; Gaucherel & Grimaldi, 2015; Wang et al., 2022). This method capitalizes on the specific acoustic patterns generated by rainfall, which vary depending on its intensity (Supplementary Figure S1 in Supporting Information S1). Previous works have mostly focused on studying this topic in urban environments (Guo et al., 2018; Wang et al., 2022), or using a limited dataset of recordings obtained below a rainforest canopy (Bedoya et al., 2017).

The Amazon basin provides a great test case for exploring innovative rainfall measurements, not only due to its remoteness and lack of road networks but also the dense tree coverage that prevents the use of conventional rain gauges. Prior research has shown the feasibility of estimating rainfall intensity using sound, our work stands out as the first to train and test a model in two distinct locations comprising three different vegetation types, thus proving the practicality and reliability of this approach across multiple sites. Our approach is based on machine learning classification and regression algorithms, while previous works have examined the possibility of identifying rainfall through its sound using data from the Amazon region (Erbs et al., 2023; Metcalf et al., 2020), our study is the first to calculate actual rainfall rate.

2. Materials and Methods

2.1. Study Area

Data was collected in two separate locations, approximately 30 km apart, with three distinct vegetation types (Figure 1).

- Location 1:
 - An area of secondary tropical rainforest within the campus of the Mamirauá Institute for Sustainable Development (IDSM), located in the city of Tefé, State of Amazonas (Brazil). The city is around 550 km west of the state capital, Manaus. The yearly average rainfall in this region is approximately 2,500 mm, the rainiest months occurring between December and April (Reboita et al., 2010);
- Location 2:
 - Low várzea: floodplain forest habitat within the Amazon, inside the Mamirauá Sustainable Development Reserve (MDSR), characterized by a high individual density of tree species but with lower size and diversity when compared to the high várzea (Ferreira-Ferreira et al., 2015);
 - Chavascal: habitat within the Amazonian wetlands and inside the MDSR, mainly composed of dense and species-poor shrubs/trees (Ferreira-Ferreira et al., 2015).

2.2. Sound Recording

We deployed three autonomous sound recorders (model Song Meter Micro, Wildlife Acoustics, Maynard, MA, USA) with build-in omnidirectional microphones (sensitivity: +2 dB FS $\pm$ 4 dB; Signal-to-Noise Ratio: 73 dB). Similar devices have been extensively used to monitor wildlife in tropical environments (Bedoya et al., 2017; Sugai et al., 2019). The data collected inside IDSM and MDSR were recorded in wav format (16 bits), a sampling rate of 48 kHz for the recorder inside the MDSR and of 24 kHz for the two recorders inside the IDSM (to increase battery life). Recorders registered a varying amount of sound data:

- **Recorder 1 (Inside IDSM):** located at 3.353,256°S, 64.733,703°W, collected approximately 799 hr, equivalent to 47,925 1 min recordings, between February 14 and October 11 of 2023;

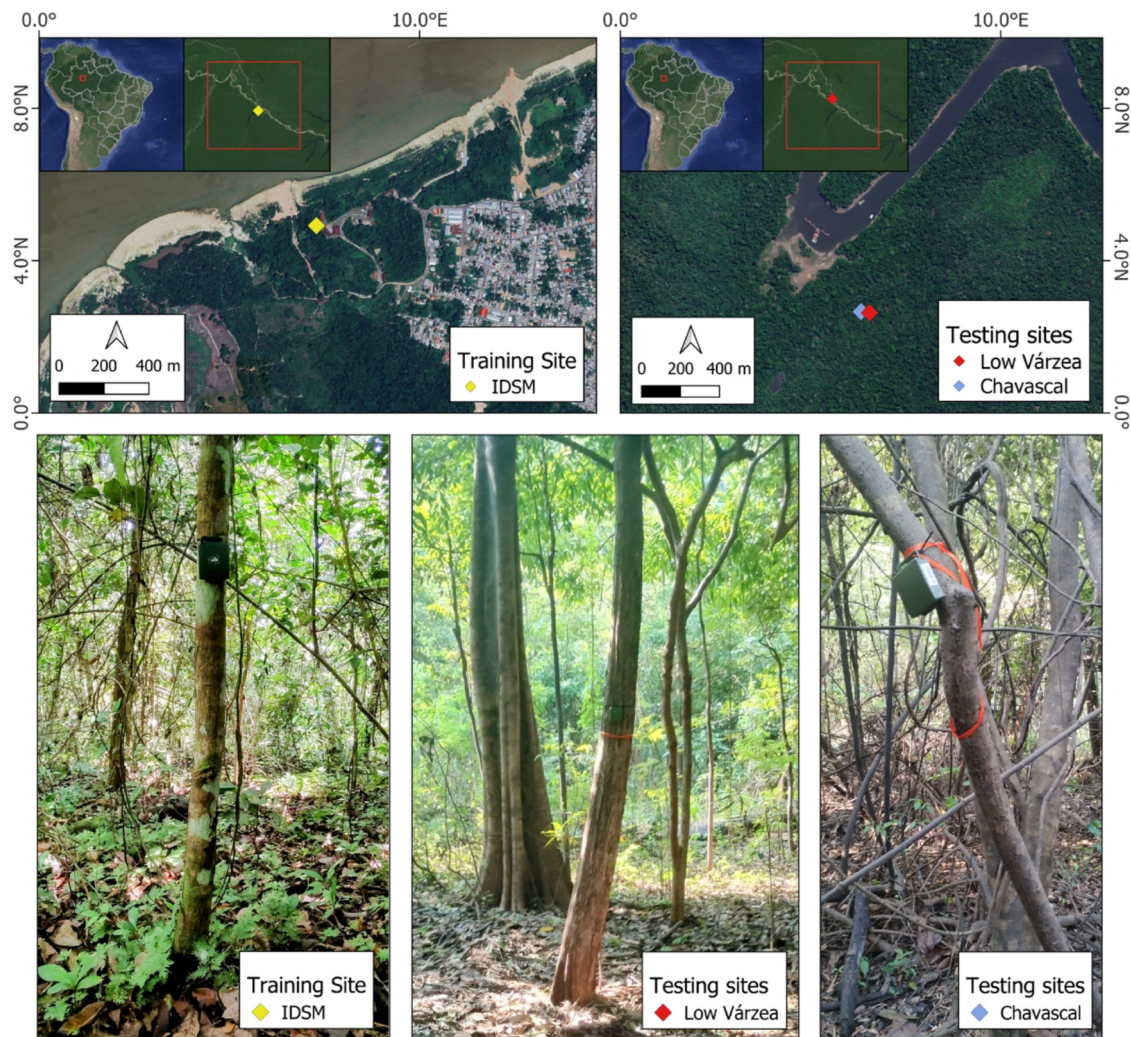


Figure 1. Set of maps depicting the area of study. The training site (top left), located at the IDSM, testing sites, located at the MSDR (top right). The bottom pictures show the vegetation around recorder 1 (left), recorder 2 (center), and recorder 3 (right). The Low Várzea and Chavascal sites are periodically flooded (located within the Amazon River floodplain).

- **Recorder 2 (Low várzea):** located at 3.066623°S, 64.846,540°W, recorded 105 hr, equivalent to 6,309 1 min recordings, obtained between December 11 of 2023 and January 20 of 2024;
- **Recorder 3 (Chavascal):** located at 3.06657°S, 64.84693°W, recorded 50 hr, equivalent to 3,027 1 min recordings, obtained between January 12-23 of 2024.

2.3. Rain Gauge Measurements

Rainfall data was collected with a tipping bucket rain gauge (model pluviDB, DualBase, Palhoça, SC, Brazil) with a precision of 0.2 mm every 5 min. Initially, the rain gauge was set within the perimeter of the IDSM in an open field area 1 m above ground, around 15 m distance from recorder 1. The same rain gauge was repositioned inside the MSDR alongside the Japurá River on the top of a floating research station operated by IDSM, approximately 500 m away from recorders 2 and 3 to be used as ground-truth for the testing data. Rainfall rates were labeled according to the World Meteorological Organization Criteria (WMO-World Meteorological Organization, 2021): light (less than or equal to 2.5 mm/hr), moderate (between 2.5 mm/hr and 10.0 mm/hr), heavy (between 10.0 mm/hr and 50.0 mm/hr) and violent rainfall (greater than 50.0 mm/hr). A summary of the rainfall data collection for each location is presented in Table 1.

Table 1
Rainfall Data Collected by Each Recorder

Recorder	Timesteps (1 min)					Total rainfall (mm)
	No rain	Light	Moderate	Heavy	Violent	
1	46,842	494	570	239	63	1,337.6
2	607	114	63	51	10	225
3	2,899	48	38	38	4	130.2

2.4. Sound Data Analysis

The analysis of the mean Power Spectrum Density (PSD) is effective in distinguishing dry and rainy periods with different intensities as they exhibit different patterns (Bedoya et al., 2017; Erbs et al., 2023; Metcalf et al., 2020). Here we use Welch's method, available in the Signal module from the Scipy library (Virtanen et al., 2020), for estimating the PSD for each one-minute recording.

2.5. Optimal Frequency Range Selection

The frequency range of 600–1,200 Hz selected by Bedoya et al. (2017) did not produce high Pearson correlation values using our dataset. Therefore, we recalculated the PSD using Welch's method with a different number of samples per segment parameter (nperseg) set with the value 1,024, which yielded better results. We moved a 600 Hz range window from 0 to 24 kHz and analyzed the correlation with the amount of rainfall (supplementary Figure S2 in Supporting Information S1). The window within the range 600–1,200 yielded low correlation values (around 0.4) and the only two frequency windows with correlations above 0.6 were 0–797 Hz (correlation of 0.72) and 1,641–2,719 Hz (0.67). To evaluate the training model's performance, we tested it using data from recorder 1 with (a) both windows combined and (b) each window separately. The range 1,641–2,719 Hz was the most promising one. The combined use of both ranges did not significantly improve results — Probability of Detection (POD) and False Alarm Rate (FAR) metrics remained the same (see Section 2.6) — but doubled the time spent in training the model. While the 0–797 Hz range could lead to model overfitting, the 1,641–2,719 Hz wider range encompasses a broader spectrum of features, making the model more adaptable to different locations, and was selected for algorithm development.

2.6. Rainfall Detection

For the detection of rainfall, we selected three different classification models: Logistic Regression (LogR), Random Forest (RF), and Gradient Boosting (GB). LogR is a straightforward and interpretable model commonly used as a baseline or reference point in machine learning tasks and was considered a benchmark against the two more complex models (RF and GB). Both RF and GB are ensemble methods, combining multiple decision trees in their structure (in a parallel manner in RF and a serial manner in GB). They are among the most powerful and widely used machine learning tools for supervised learning (Müller, 2017).

The original data of recorder 1 was split into training (75%) and validation (25%) subsets. The classifiers were trained with an equal amount of “rain” and randomly selected “no rain” data, avoiding the risk of bias toward the majority class. One hundred subsets for training and validation were sampled, minimizing the risk of removing important data or introducing meaningless information during the random selection of “no rain” audio data (Krawczyk, 2016). After training and validating the models one hundred times, we calculated the mean Probability of Detection (POD) and False Alarm Rate (FAR) according to Equations 1 and 2 below:

$$POD = \frac{TP}{TP + FN} \quad (1)$$

$$FAR = \frac{FP}{FP + TN} \quad (2)$$

where TP and FP are the number of true and false positives, respectively, and TN and FN the number of true and false negatives. The model with best POD and FAR was then selected for the final structure of the algorithm. All machine learning algorithms in this study were implemented using the sklearn library available in Python (Pedregosa et al., 2011), and the parameters were defined with the hyperparameter optimization scheme provided by the GridSearchCV module.

2.7. Rainfall Estimation

We selected three models to estimate rainfall rates: Linear Regression (LR), RF, and GB, where LR plays the same benchmark role as the LogR in Section 2.6. The input parameter for LR consisted of the integrated PSD

within the best frequency band. Models were trained using only data from rainy periods, following the rainfall detection obtained from the process described in Section 2.6.

2.8. Validation and Final Algorithm Structure

After performing the rainfall detection test described in Section 2.6 and selecting one of the classification models, the final algorithm consisted of the following steps.

1. Removal of 10 days from the dataset, to be used as validation;
2. Train the best classifier based on Section 2.6 results, with an equal amount of rain/no rain data;
3. Train LR, RF, and GB regression models with only rain data;
4. Apply the classifier on the validation dataset;
 - 4.1. If rainfall is detected by the classifier, predict rainfall values with the regression models;
 - 4.2. If no rainfall is detected, the estimated rainfall is zero;
5. Return the 10-day period to the original dataset and take the next 10 days;
6. Repeat training and validation steps until every available 10-days period is tested;
7. Calculate POD/FAR for the classification models and R^2 for the regression models considering only rainfall values greater than zero;
8. Retrain the best classifier from Section 2.6 and the best regression model from Step 7 (based on R^2 values), using the complete dataset from recorder 1, testing it on the datasets from recorders 2 and 3;
9. Evaluate the model's performance on data from recorders 2 and 3 (POD and R^2); comparison of total rainfall estimated by the model and daily satellite data from the Global Precipitation Measurement (GPM), with a spatial resolution of 0.1° (Huffman et al., 2015).

3. Results and Discussion

3.1. Model Training and Validation at Site 1

Classification/regression tests on the training site (IDSMSite) and classification/regression on test sites using data from recorder 1 as training are detailed in this section. Figure 2 displays 60 random PSD samples for each rainfall category (in order of intensity) from recorder 1, which shows the relationship between rainfall intensity and power spectrum values.

The dataset utilized to train and test the classifiers using only data from recorder 1 consisted of a total of 2,732 recordings. The dataset was split equally between “rain” and “no rain” categories. Overall, in this first classification test all three models performed well on the task of identifying rainfall events, with RF achieving the highest POD (0.98) followed closely by GB (0.97). The LogR model yielded the lowest number of false positives (FAR of 0.02), while both RF and GB yielded the same FAR of 0.04. Results from the initial classification test demonstrated that RF and GB are equally efficient for detecting rainfall events, therefore, in the final algorithm only the RF classifier is utilized, due to its shorter running time. The regression models (RF and GB) yielded a coefficient of determination (R^2) of 0.62 and 0.56, respectively. The models' performances showed significant improvement when considering a longer integration time (Figure 3). For instance, when comparing the accumulated rainfall over a 60-minute period, both RF and GB presented R^2 values higher than 0.85. This improvement arises from the fact that some model estimates are not always perfectly synchronized with the rain gauge measurements (Supplementary Figure S4 in Supporting Information S1). However, all models underestimated the highest rainfall rates, although this may be related to the reduced number of samples for this kind of event.

The rain gauge recorded a cumulative rainfall of 1,337.6 mm between February 14 and October 11 during the data collection of recorder 1. Utilizing integrated PSD values from sound recordings, the LR model estimated a total rainfall of 782.5 mm, while RF predicted 1,384.3 mm and, lastly, GB estimated 1,272.8 mm.

After performing steps 1 to 7 (see Methods Section 2.8), the RF classifier achieved a POD of 0.88 and FAR of 0.03. The R^2 scores of the rainfall time series are very high, above 0.87 for hourly rain rates (Figure 3a). The underestimation of the highest rain rates can be attributed to either a technical limitation of the employed microphone (e.g., signal saturation) or the scarcity of such events in the training data set.

Regarding the detection of rainfall/no-rainfall events, the performance metrics are in agreement with Metcalf et al. (2020), who tested an algorithm to detect rainfall in several locations, reaching a POD of 86% for a dataset

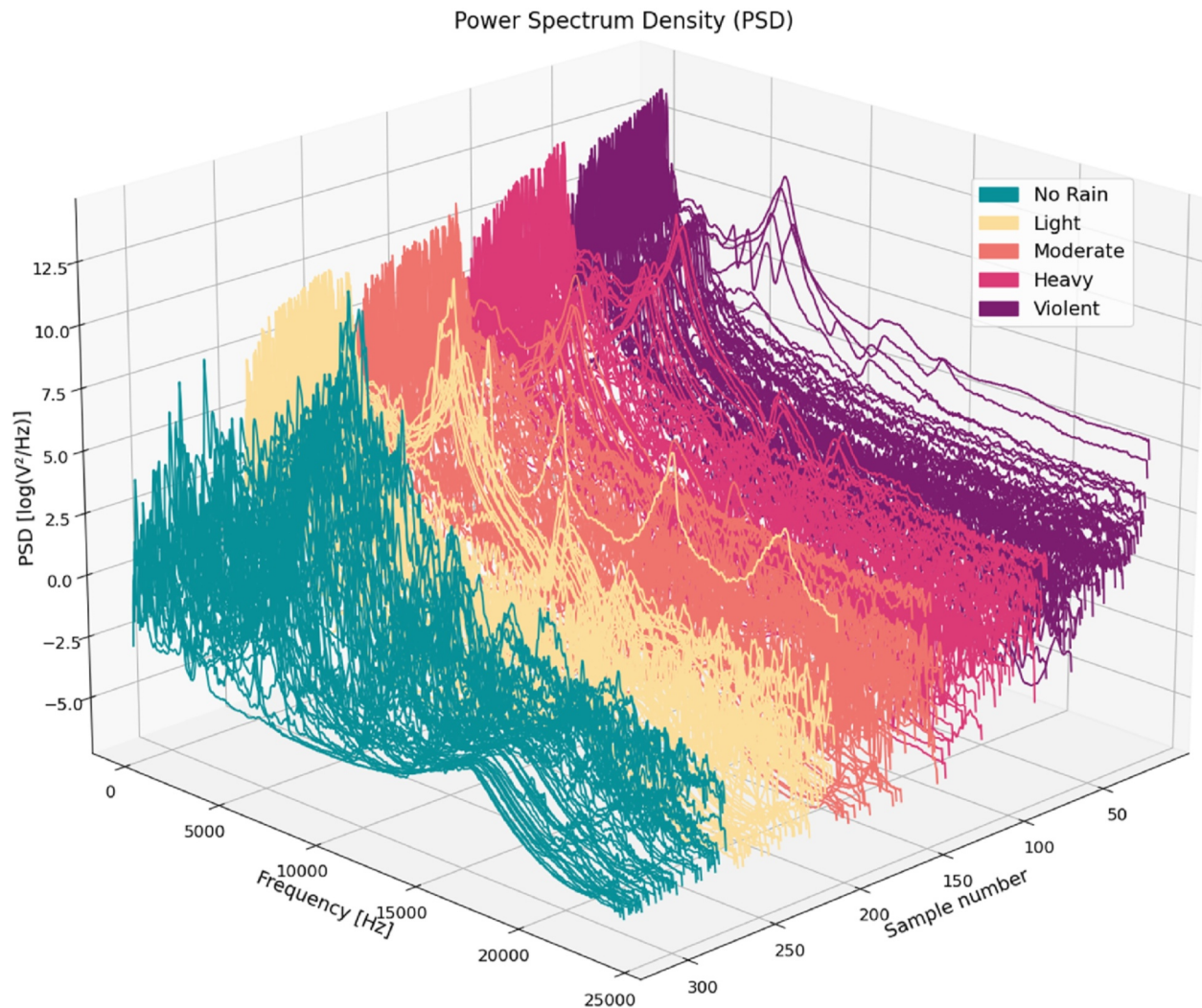


Figure 2. Power Spectrum Density of 60 random samples from recorder 1 for each rainfall category (no rain, light, moderate, heavy, and violent rain). Overall, the more extreme the rainfall, the higher the PSD value.

collected in a tropical rainforest in Santarém, Pará State, Brazil. However, our metric values are comparatively lower than those documented in Bedoya et al. (2017), who achieved a POD of 92% for classifying rainfall sounds in a Colombian rainforest. Additionally, Erbs et al. (2023) developed a Convolutional Neural Network for river dolphin monitoring inside the MDSR, which achieved an average precision of 0.98 for the rain class.

3.2. Model Application to Other Amazon Forest Sites

The validation process indicates that the RF model is the most suited for generalization (Figure 3a). Therefore, for the final test using data from recorders 2 and 3, we trained an RF classifier/regressor using only data from recorder 1 (step 8 of Section 2.8). Our model was capable of identifying rainfall with a 97% accuracy for the Low várzea environment and 98% for Chavascal. Regression results for the Low várzea and Chavascal environments yielded an R^2 of 0.69 and 0.93 for hourly rainfall rates (Figure 3b).

During the data collection period, the rain gauge measured a total of 225 mm (130.2 mm) for the Low várzea (Chavascal) area, while the RF model estimated a total precipitation of 242.3 mm (112.4 mm), showcasing its accuracy in estimating rainfall amounts. To put these results in perspective with other datasets available, we included in our comparisons the Global Precipitation Measurement (GPM) satellite constellation IMERG-Final product. IMERG estimated a total rainfall of 348.7 mm over Low várzea and 79.6 mm over Chavascal during the

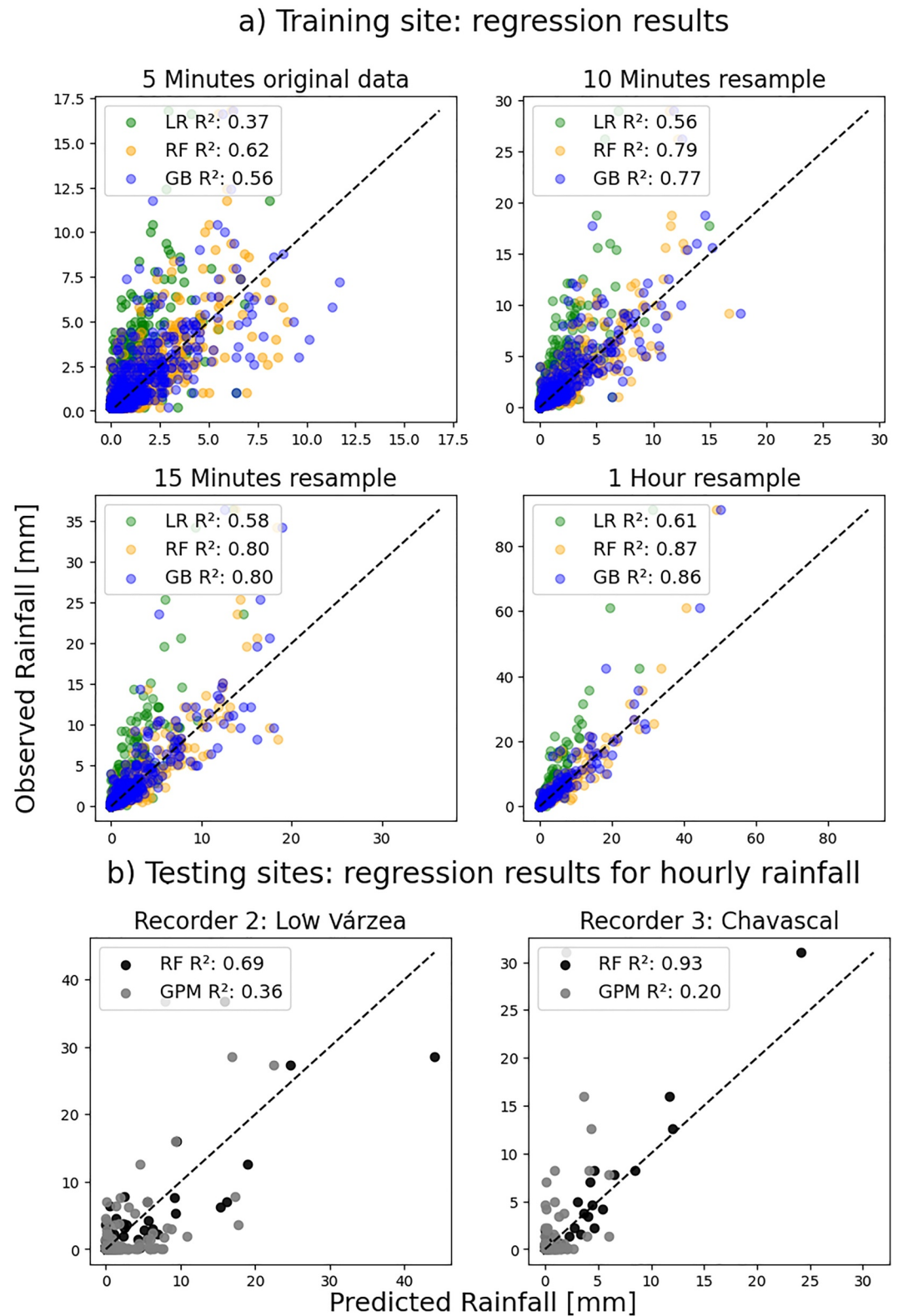


Figure 3. Scatter plots of estimated (x-axis) versus gauge reference (y-axis) rainfall values, (a) for the entire validation dataset and different integration time and (b) scatter plots of hourly rainfall values for recorders 2 and 3 estimated using the model trained with data from recorder 1; the estimates from the GPM IMERG Final satellite product are superimposed in gray, for comparison.

assessed period. Root Mean Square Error (RMSE) between the rain gauge and the RF model was 1.33 mm/hr for Low Várzea and 0.68 mm/hr for Chavascal. In contrast, the RMSE between the rain gauge and the GPM estimates is significantly higher, at 2.03 mm/hr for Low Várzea and 2.3 mm/hr for Chavascal. The scatter plots in Figure 3b further illustrate the agreement between the RF model and the rain gauge, and the discrepancy when compared to the satellite product. These results corroborate the relevance of combining sound information with an RF model to estimate high-resolution rainfall rates.

Our results confirm the relevance of using soundscape to estimate rainfall rate at high temporal resolution in remote areas poorly covered by rain gauge networks. Although our results provide great perspectives for future large-scale monitoring of rainfall dynamics in remote areas, challenges such as data collection, storage, and battery management pose limitations that must be overcome. Nevertheless, decreasing the recording time or the sampling frequency could be investigated. A possible solution would be to invest in real-time data transfer such as proposed in the recent Providence Project (Erbs et al., 2023). Another limitation of the current technique is the use of supervised classification and a nearby rain gauge was needed to train the machine learning models. Our regression results show how powerful a predictor based on PSD can be, whilst promising, the testing data sample pool of recorders 2 and 3 lack violent rainfall events, which our tests with recorder 1 demonstrated to be the source of most errors during the regression task. High-resolution, 5-minute rainfall data located in different locations would help characterize rainfall variability at scales which cannot be captured by satellite products. Additionally, the use of sound recorders allows for the detection of the onset and duration of rainfall events with increased precision (Dunkerley, 2020). This data can also be of interest in other fields of study such as ecology, as rainfall plays a major role in behavioral and populational models. Future work will continue to address generalization to other locations and how the inclusion of additional input features impacts the model's performance. Moreover, to address the underestimation of high rainfall rates we will continue to collect training data, improving the models' performance in future developments, and utilizing different dB gains with the goal of better understanding the acoustic signal of violent rainfall events. Additional research will be carried out regarding methods to better find and implement the optimal frequency range.

4. Conclusions

Historically, geophonic sounds such as rain and wind have been overlooked in soundscape ecology, viewed more as background noise than valuable data. However, recent advancements in automated analysis have shed light on their potential for rainfall identification. Our study expanded upon this concept, by training a machine learning model with an extensive dataset based solely on PSD values from the frequency range 1,641–2,719 Hz of sound recordings inside a secondary tropical forest. To our knowledge, for the first time, a model trained at one site with rainfall sound data is validated over two other sites (and in our case, different vegetation types), over 30 km of distance from the training site. Our approach reached over 97% accuracy for detecting rainfall and showed a high potential for estimating hourly rainfall rates of light, moderate, and heavy intensities. The model's performance surpassed satellite data estimates, as evidenced by ground-truth observations from a rain gauge.

The sound of Amazon rainfall can be used to monitor this essential climate variable, fostering climate monitoring and disaster management in the region but also other tropical areas across the globe. This is an important result, and our study opens the way to a new opportunistic approach for rainfall monitoring in a poorly gauged region due to its remoteness and high tree cover. The increased interest in biodiversity monitoring and the development of low-cost and high-quality acoustic sensors is resulting in several passive acoustic monitoring networks being deployed across the tropical region (Darras et al., 2024). Soon, they could be used for large-scale, high-resolution monitoring of rainfall, creating a bridge between climate and biodiversity monitoring.

Data Availability Statement

All developed codes, datasets and supplementary material are available at Xavier et al., 2023.

Acknowledgments

There is not any relationship between the manufacturer of the recorder and rain gauge used for this research and the authors of this paper. The authors acknowledge the funding by Gordon and Betty Moore Foundation (Project “Advancing the understanding of methane emissions from tropical wetlands”), the Institut de recherche pour le developpement (IRD), “Fundação de Amparo à Pesquisa do Estado do Amazonas” (FAPEAM), “Secretaria de Estado de Desenvolvimento Econômico, Ciência, Tecnologia e Inovação” (SEDECTI) and the “Governo do Estado do Amazonas” – EDITAL 003/2023 - PAINTER INFRA CT&I, and the “Fundo Nacional de Desenvolvimento Científico e Tecnológico” (FNDCT). We thank the support from the Amazonas State Secretariat for the Environment (SEMA).

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